

Digital Logic DesignClass-3

23/06/2026

Logic 0 : $0 \sim 0.8 \text{ V}$
 Logic 1 : $2.4 \sim 5.0 \text{ V}$ } International Standard

- Analog system processes physical quantities that exist in a continuous range of values, where signals vary smoothly without distinct steps.
- Digital system consists of devices that process logical information or physical quantities in digital form, where signals can assume only discrete and separate values rather than continuous ones.

Difference between analog and digital system?

Feature	Analog	Digital
How it looks	Smooth, continuous waves.	Sharp, stepped jumps (like blocks).
Values	Infinite choices (e.g., any fraction).	Fixed choices (usually just 0 and 1, or On and Off).
Examples	a mercury thermometer, a wall clock with moving hands.	a smartphone screen, a digital watch.
Static & Noise	Low noise immunity	High noise immunity
Storage	Hard to save perfectly without it wearing out over time.	Very easy to save, copy, and send without any changes.

Common Number System (Decimal, Binary, Octal, Hexadecimal)

Why use octal and Hexadecimal?

⇒ because we can perform bitwise manipulation in case of octal and Hexadecimal.

If we take $(127)_{10}$, we can't convert it to binary bitwise

But if we take $(127)_8$, we can convert it like this:

$(1 \ 2 \ 7)_8$
 $\swarrow \quad \downarrow \quad \searrow$
 $(001 \ 010 \ 111)_2$

$(1 \ 2 \ 7)_{16}$
 $\swarrow \quad \downarrow \quad \searrow$
 $(0001 \ 0010 \ 0111)_2$

Notewise

Conversion

Decimal to Other: (Base নির্দিষ্ট ভাগ/গুন)

$$(232.16)_{10} = (?)_8$$

$$\begin{array}{r} 8 \overline{) 232} \\ 8 \overline{) 29} - 0 \\ 8 \overline{) 3} - 5 \\ 0 - 3 \end{array}$$

$$\therefore (232)_{10} = (350)_8$$

$$\therefore (.16)_{10} = (.1217...)_{8}$$

$$\therefore (232.16)_{10} = (350.1217)_8$$

$$\begin{array}{r} 0.16 \\ \times 8 \\ \hline 1.28 \\ 0.28 \\ \times 8 \\ \hline 2.24 \\ 0.24 \\ \times 8 \\ \hline 1.92 \\ 0.92 \\ \times 8 \\ \hline 7.36 \end{array}$$

Binary to Octal:

ডানপাশ থেকে 3 টা করে Group.

$$\begin{array}{ccccccc} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ \hline & & 1 & 4 & 6 & 7 & & & & & & \end{array}$$

$$\begin{array}{l} 1 \text{ K} = 2^{10} (1024) \\ 1 \text{ M} = 2^{20} \\ 1 \text{ G} = 2^{30} \end{array}$$

Other to Decimal:

$$\sum \text{value} \times \text{Base}^{\text{weight}} \rightarrow \text{Position}$$

$$(41C.D9)_{16} = (?)_{10}$$

$$= 4 \times 16^2 + 1 \times 16^1 + 12 \times 16^0 + 13 \times 16^{-1} + 9 \times 16^{-2}$$

$$= (1052.848)_{10}$$

Binary to Decimal: (Shortcut)

$$(1001110)_2 = (?)_{10}$$

$$\begin{array}{ccccccc} 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ \textcircled{64} & 32 & 16 & \textcircled{8} & \textcircled{4} & \textcircled{2} & 1 \end{array}$$

$$= 64 + 8 + 4 + 2 = (78)_{10}$$

Binary to Hexadecimal:

ডানপাশ থেকে 4 টা করে Group.

$$\begin{array}{ccccccc} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ \hline & & 4 & 4 & 7 & A & C & & & & & & & & \end{array}$$

Class-4,5

28/06/2026

Number conversion system in C++

Binary Arithmetic:

- Addition
- Subtraction
- Multiplication
- Division

Complements of Binary Numbers:

- i) 1's complement (0 to 1 and 1 to 0 conversion)
- ii) 2's complement

$$101011 \xrightarrow{1's} 010100 \xrightarrow[+1]{2's} 010101$$

Shortcut:

LSB থেকে First 1 আসা পর্যন্ত same থাকবে। পরবর্তীগুলো 1's complement.

$$\begin{array}{c} \text{MSB} \quad \text{LSB} \\ \underline{10100} \end{array} \longrightarrow \begin{array}{c} \underline{01100} \\ \text{1's complement} \quad \text{Same} \end{array}$$

Signed Numbers

MSB (Left most bit) - sign বোঝায়।

$$0 \rightarrow (+)$$

$$1 \rightarrow (-)$$

Range of values

- Total combination = 2^n
- Unsigned numbers: 0 to $(2^n - 1)$
- 2's complement: (-2^{n-1}) to $(2^{n-1} - 1)$
- Sign Magnitude / 1's complement: $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$

For 8 bit numbers:

$$\text{Total combination} = 2^8 = 256$$

$$\begin{aligned} \text{Range: } & -2^{8-1} \text{ to } 2^{8-1} - 1 \\ & = -128 \text{ to } 127 \end{aligned}$$

Multiplication:

- Both numbers must be in uncomplemented form
- Multiply 3 by -5.

Opposite signs, so product will be negative.

$$\begin{aligned} 3_{10} &= 00000011_2 \\ -5_{10} &= 11111011_2 \end{aligned}$$

00000011	Multiplcand
<u>X 0000101</u>	Multiplier
00000011	First partial product
<u>+ 0000000</u>	Second partial product
00000011	Sum of 1 st and 2 nd
<u>+ 000011</u>	Third partial product
00001111	Sum and Final Product

Final result is negative, so take 2's complement.
11110001 is the result which in decimal is -15.

Why 1 negative less in sign magnitude?
Because of +0 and -0 being separately counted in sign magnitude.

Division: Division is equivalent to subtracting divisor from dividend a number of times equal to the quotient.

- Both numbers must be in uncomplemented form
- Divide 01100100 by 00110010.

Both numbers are positive so quotient will be positive.
Set the quotient to zero initially.

Subtract the divisor from the dividend by using 2's complement addition. (11001110)
Ignore the carry bit.

Subtract the divisor from the 1st partial remainder using 2's complement addition.

01100100	Dividend
+ 11001110	2's complement of Divisor
1 00110010	First partial remainder
Add 1 to quotient: 00000000 + 1 = 00000001	
00110010	First partial remainder
+ 11001110	2's complement of Divisor
1 00000000	zero remainder
Add 1 to quotient: 00000001 + 1 = 00000010	

So final quotient is 00000010 and final remainder is 00000000

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BCD Code: BCD is not a number system. It is a decimal number with each digit encoded to its binary equivalent (4-bit)

$$\begin{aligned} (13)_{10} &= (1101)_2 \\ \swarrow \quad \downarrow \\ 0001 \quad 0011 \\ &= (00010011)_{BCD} \\ &= (10011)_{BCD} \end{aligned}$$

- BCD addition - a correction factor = $(6)_{10} = (0110)_2$

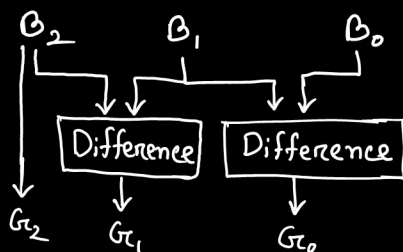
$$\begin{array}{r} 35 = (00110101)_{BCD} \\ 28 = (00101000)_{BCD} \\ \hline 63 = 01011101 \\ \quad \quad \quad + 0110 \\ \hline 01100011 \\ \quad \quad \quad \underbrace{\quad\quad\quad}_6 \quad \underbrace{\quad\quad\quad}_3 \end{array}$$

যেহা (কোনো একটি) binary-এর decimal equivalent value > 9 তাহলে, আমরা সংশোধন $(0110)_2$ যোগ করতে হবে।

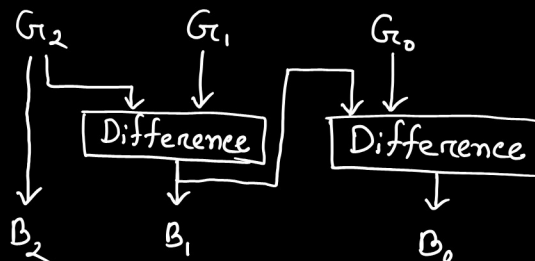
$$\begin{array}{r} 75 = (01110101)_{BCD} \\ 68 = (01101000)_{BCD} \\ \hline 143 = 11011101 \\ \quad \quad \quad (+) \boxed{0110} \boxed{0110} \\ \hline 000101000011 \\ \quad \quad \quad \underbrace{\quad\quad}_1 \quad \underbrace{\quad\quad}_4 \quad \underbrace{\quad\quad}_3 \end{array}$$

Gray code:

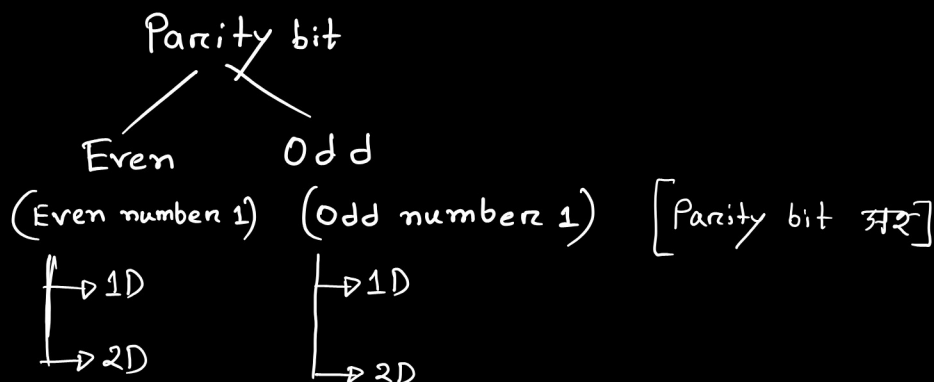
Binary to Gray



Gray to Binary



Parity Method for Error Detection:

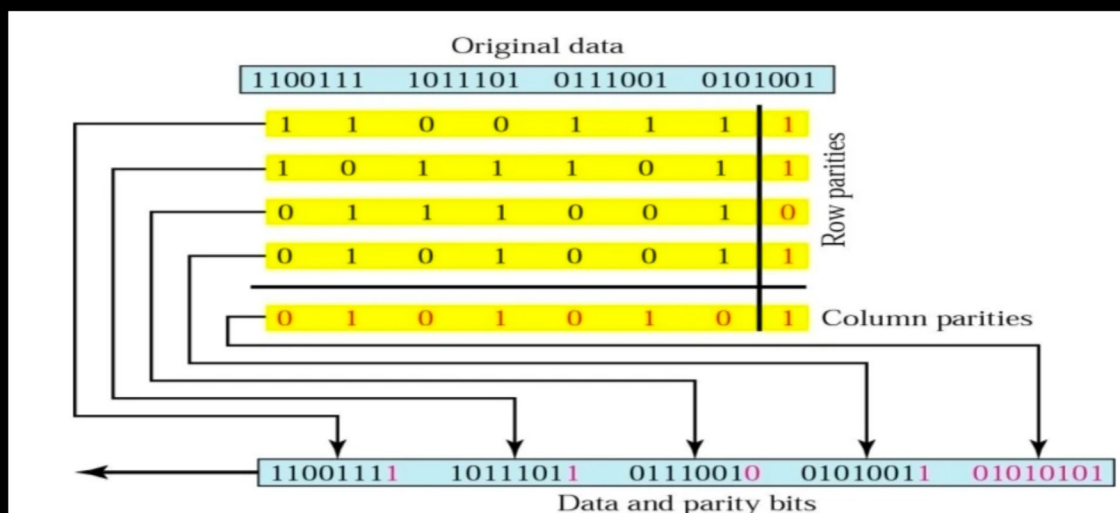


- Generally, MSB-70 Parity bit સમાવેશ થાય છે.
- Generally, even parity use થાય છે.

1D Parity checking / Simple Parity check :

It can only detect single bit error or odd number of errors.

2D Parity checking:



Two- Dimension Parity Checking increases the likelihood of detecting burst errors.

Class-7,8

Class-9

Class-10

5/7/26

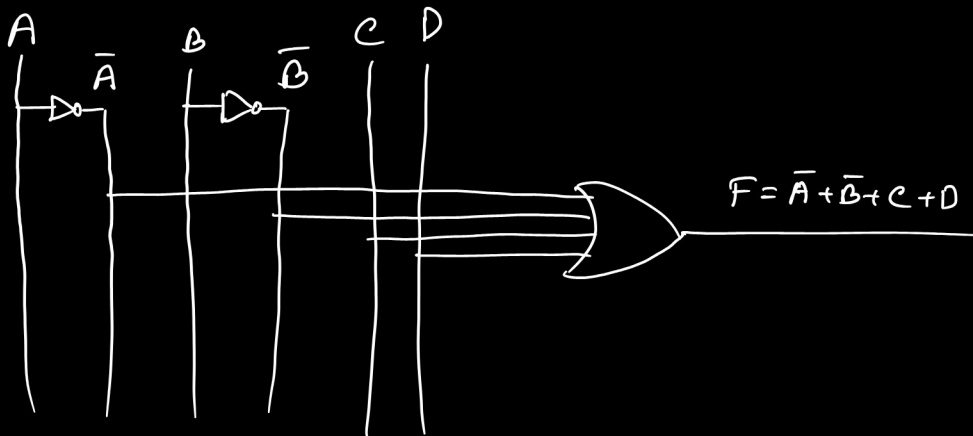
11/07/26

12/07/26

$$F = \overline{AB(\overline{C+D})}$$

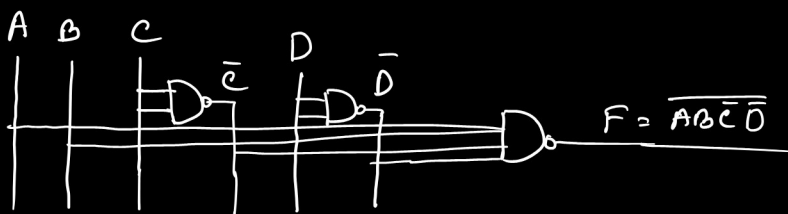
$$= \overline{AB} + \overline{(\overline{C+D})}$$

$$= \overline{A} + \overline{B} + C + D$$



$$F = \overline{AB(\overline{C+D})} \rightarrow \overline{AB\overline{C+D}}$$

$$= \overline{AB\overline{C}\overline{D}}$$



Difference between hypothesis, theorem and formula.

$$F(A, B, C) = \Sigma(0, 1, 2, 3, 6, 7)$$

$$= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + AB\overline{C}$$

$$= \overline{A}\overline{B}(C + \overline{C}) + B\overline{C}(A + \overline{A}) + C\overline{C}(A + \overline{A})$$

$$= \overline{A}\overline{B} + B\overline{C} + C\overline{C}$$

0 → 000
1 → 001
2 → 010
3 → 011
6 → 110
7 → 111

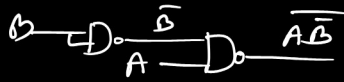
Notewise

$$= \bar{A}\bar{B} + B(c + \bar{c})$$

$$= \bar{A}\bar{B} + B$$

$$= \bar{A} + B$$

$$\begin{aligned} & \overline{\overline{A+B}} \\ &= \overline{\bar{A} \cdot \bar{B}} \\ &= \overline{\bar{A}\bar{B}} \end{aligned}$$



Class - 11, 12

18/07/26

X-OR

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0



$$F(A, B) = \Sigma(1, 2)$$

$$= \bar{A}B + A\bar{B}$$

$$= A \oplus B$$

- X-OR is used as comparator or parity generator.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

$$F(A, B, C) = \Sigma(4, 5, 7)$$

$$\Rightarrow F = A\bar{B}\bar{C} + A\bar{B}C + ABC$$

$$= A\bar{B}(C + \bar{C}) + ABC$$

$$= A\bar{B} + ABC$$

$$= A(\bar{B} + BC)$$

$$= A(B + \bar{B})(\bar{B} + C)$$

$$= A(\bar{B} + C)$$

$$A\bar{B} + AC$$

$$A\bar{B}C + A\bar{B}\bar{C} + ABC$$

$$A \vee (A \wedge B) \equiv (A \wedge T) \vee (A \wedge B)$$

$$A + AB \equiv A \wedge (T \vee B)$$

$$\equiv A \wedge T$$

$$= A$$

$$A + \bar{A}B \equiv (A + \bar{A})(A + B)$$

$$\equiv (A + B)$$

Active $\rightarrow 1$
0 $\rightarrow$ $\bar{0}$

Active $\rightarrow 0$
1 $\rightarrow$ $\bar{1}$

Input		Output	SOP (Minterms)	POS (Maxterms)
A	B	AB		
0	0	0	$\bar{A}\bar{B}$	$A+B$
0	1	0	$\bar{A}B$	$A+\bar{B}$
1	0	0	$A\bar{B}$	$\bar{A}+B$
1	1	1	AB	$\bar{A}+\bar{B}$

In case of SOP, $F = AB$

$$\begin{aligned}
 \text{" " " POS, } F &= (A+B)(A+\bar{B})(\bar{A}+B) \\
 &= (A+B)(A\bar{A}+A\bar{B}+\bar{A}B+B\bar{B}) \\
 &= (A+B)(AB+\bar{A}B) \\
 &= AAB + A\bar{A}B + ABB + \bar{A}B\bar{B} \\
 &= AB + AB \\
 &= AB
 \end{aligned}$$

#

$$F = A + BC$$

$$= A \cdot 1 + BC \cdot 1$$

$$= A(B + \bar{B}) + BC(A + \bar{A})$$

$$= AB + A\bar{B} + ABC + \bar{A}BC$$

$$= AB(C + \bar{C}) + A\bar{B}(C + \bar{C}) + ABC + \bar{A}BC$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC + \bar{A}BC$$

$$F = ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}BC \quad \rightarrow \text{Canonical Form}$$

$$F = \sum (3, 4, 5, 6, 7)$$

Again, $F = A + BC$

$$= (A+B)(A+C)$$

$$= (A+B+0)(A+0+C)$$

$$= (A+B+C\bar{C})(A+B\bar{B}+C)$$

$$= (A+B+C)(A+B+\bar{C})(A+B+C)(A+\bar{B}+C)$$

$$\Rightarrow F = (A+B+C)(A+B+\bar{C})(A+\bar{B}+C) \quad \rightarrow \text{POS}$$

$$\therefore F = \prod (0, 1, 2)$$

Distribution Law
 $(A+B)(A+\bar{B}) = A + B\bar{B}$

$$F = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C}$$

$$= \bar{A}\bar{B} + \bar{A}B + AB\bar{C}$$

$$= \bar{A} + AB\bar{C}$$

$$= (\bar{A} + B)(\bar{A} + \bar{C})$$

$$= \bar{A} + B\bar{C}$$

$$G_L = (\bar{A} + B + C)(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + \bar{C})$$

A	B	C	F	G _L
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0

Class-14,15

26/07/26

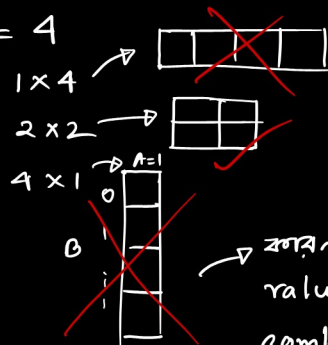
Karnough Maps (K-map)

- Only provides simple solution with basic gates. Cannot provide complex solution. It is a limitation of k-map.

Two variable k-map

$$\text{cell সংখ্যা} = 2^n \quad [n = \text{variable সংখ্যা}]$$

In case of 2-variables, cell = $2^2 = 4$



১৪৪৪ এখানে column একটিই। তাই A-এর value শুধু ০ বা ১-ই বসানো যায়। যেভাবে combination হবে না।

	A (MSB)	
	0	1
B	0	1
	0	1
	00	10
	01	11

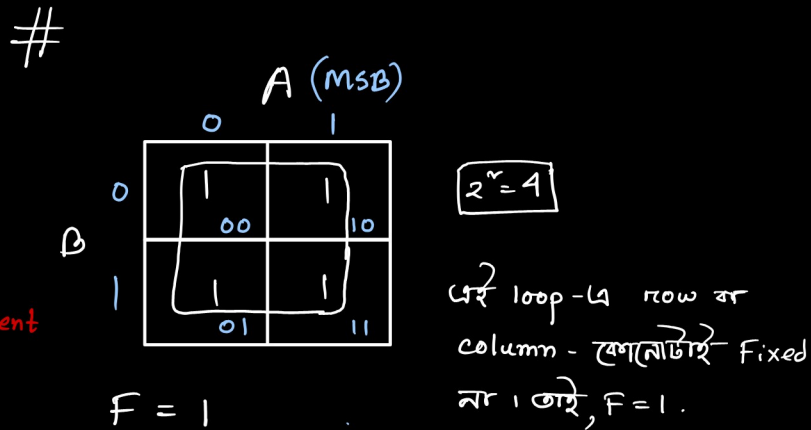
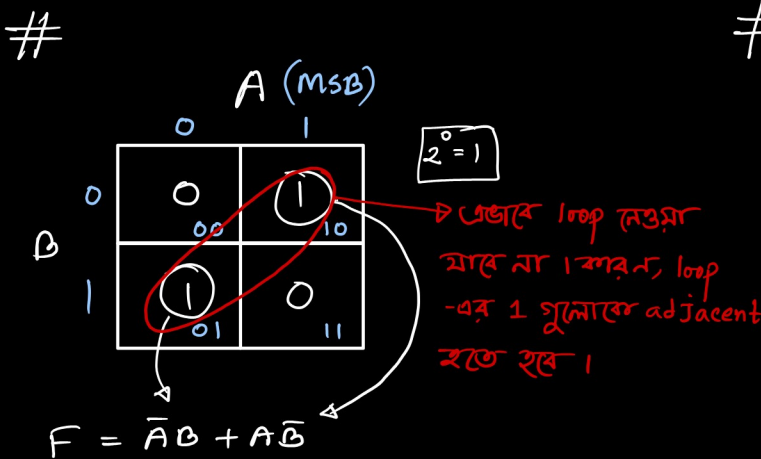
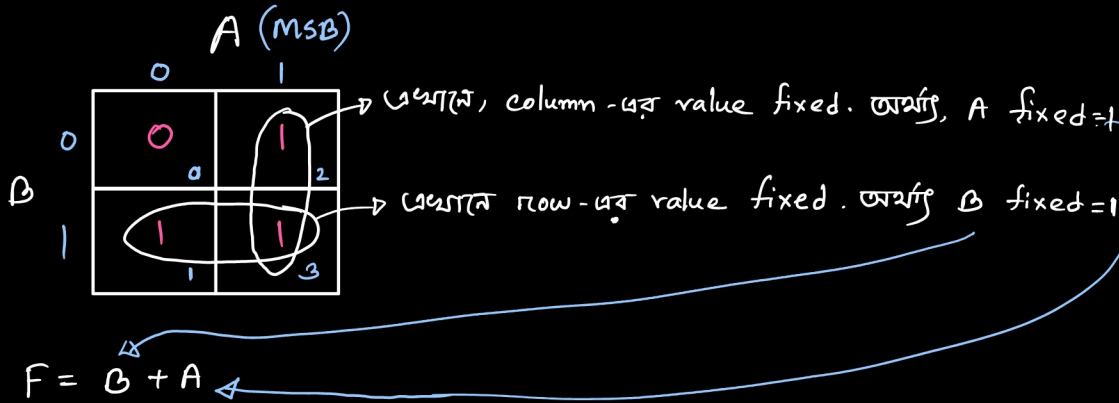
	B	
	0	1
A (MSB)	0	1
	0	1
	00	01
	10	11

	A (MSB)	
	0	1
B	0	1
	0	1
	0	2
	1	3

	B	
	0	1
A (MSB)	0	1
	0	1
	0	1
	2	3

	MSB	
	A	B
F	0	0
	0	1
	1	0
	1	1

- Loop-এ 0 থাকলে না, 1 থাকলে 2^n সূত্র অনুযায়ী। Loop-এর 1-সুন্নোকে Adjacent হতে হবে।
 $2^n \rightarrow 0, 1, \dots$ Because there are not 4 1's in this k-map
 n -এর value highest $\rightarrow$ lowest হতে হবে।
 $= 2^1 = 2$
- At least loop-এর একটি 1-কে unique হতে হবে যেন অন্য কোন loop-এ নেই।



Three-variable k-map :

Cell সংখ্যা $= 2^3 = 8$

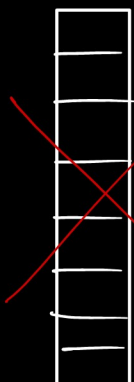
8×1

4×2

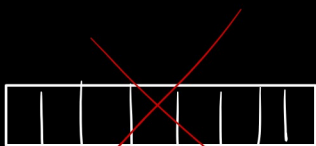
2×4

1×8

8×1 :



1×8 :



4×2 :

		A (MSB)	
		0	1
BC	00	0	1
	01	2	3
	11	6	7
	10	4	5

2×4 :

		BC			
		00	01	11	10
A (MSB)	0	0	1	3	2
	1	4	5	7	6

(MSB) AB

		BC			
		00	01	11	10
C	0	0	2	6	4
	1	1	3	7	5

✓ This will work

(MSB) AC

		BC			
		00	01	11	10
B	0				
	1				

✗ This will not work.

Continuously - এর side-এ মোত

If it were:

		A	
	0	1	
BC	00		
	01	1	
	10	1	
	11		

From k-map, $F' = \bar{A}$

$$\begin{aligned} \text{From truth table, } F &= \bar{A}\bar{B}C + \bar{A}B\bar{C} \\ &= \bar{A}(\bar{B}C + B\bar{C}) \\ &= \bar{A}(B \oplus C) \neq F' \end{aligned}$$

For this:

		A	
	0	1	
BC	00		
	01	1	
	11	1	
	10		

From k-map, $F' = \bar{A}C$

$$\begin{aligned} \text{From truth table, } F &= \bar{A}\bar{B}C + \bar{A}BC \\ &= \bar{A}C(\bar{B} + B) \\ &= \bar{A}C = F' \end{aligned}$$

That's why those 2 positions are swapped.

$F(A, B, C) = \Sigma(1, 3, 5, 2)$

		BC			
		00	01	11	10
A (msb)	0	0 ₀	1 ₁	1 ₃	0 ₂
	1	0 ₄	1 ₅	1 ₇	0 ₆

~~$2^3 = 8$~~

$2^2 = 4$ ✓

$= 4$ ✓

BC

	00	01	11	10
A	0 ₀	1 ₁	1 ₃	0 ₂
(msb)	0 ₄	1 ₅	1 ₇	0 ₆

$F = C$

#

			msb	
		AB		
	00	01	11	10
C	0	1	1	1
	1	0	0	1

$F = \bar{C} + \bar{B}$

#

		BC			
		00	01	11	10
A (msb)	0	1 ₀	1 ₁	0 ₃	1 ₂
	1	1 ₄	1 ₅	0 ₇	1 ₆

$\frac{2^3}{2} = 4$

$F = \bar{B} + \bar{C}$
 $= \bar{B}C$

#

Handwritten Karnaugh map for variables A, B, and C. The map is a 2x4 grid with columns labeled 00, 01, 11, 10 and rows labeled 0, 1. The cells contain values 1, 2, 3, 4, 5, 6, 7, 8. Orange circles highlight the 1s in the first column (A=0) and the 1s in the last column (A=1). To the right, a handwritten calculation shows $2^2 = 4$.

	00	01	11	10
0	1	2	3	4
1	5	6	7	8

$2^2 = 4$

$F = \bar{B} + \bar{C} + \bar{A}$

		BC			
		00	01	11	10
A	0	0	1	1	2
	1	4	5	7	6

		AB			
		00	01	11	10
C	0	0	1	2	4
	1	1	3	2	5

$$F = AC + BC + \bar{A}B$$

$$F = \bar{A}B + AC$$

$$= AC + BC + B\bar{A} + \bar{A}B$$

$$= AC + \bar{A}B + B(C + \bar{A})$$

=

?

$$= \bar{A}B + AC$$

		BC			
		00	01	11	10
A	0	0	1	1	2
	1	4	5	7	6

→ এটার loop নেওয়া যায় না।

$$\checkmark 2^n = 2^2 = 4$$

কিন্তু এখানে First 1 এবং Last 1 adjacent না।
তার loop রাখা না।

		BC			
		00	01	11	10
A	0	0	1	1	2
	1	4	5	7	6

→ এটার loop রাখ।

যদিও সবগুলো 1 adjacent.

Four Variable K-map :

$$\text{Cell সংখ্যা} = 2^4 = 16$$

~~16x1~~

~~2x8~~

~~4x4~~

~~8x2~~

~~1x16~~

These two are not possible because it generates invalid combination.

8x2:

A

BCD

m → minterm

M → maxterm

4x4:

	CD			
	00	01	11	10
MSB _R AB				
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

	CD			
	00	01	11	10
MSB _R AB				
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

#

	CD			
	00	01	11	10
MSB _R AB				
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

⊛ Adjacent হলে atleast - ১টি

variable remove করা। $\therefore$ mathematically

যেকোন ২টি যন্ত্র যে দুইটি element adjacent.

#

	CD			
	00	01	11	10
MSB _R AB				
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

$$\times 2^1 = 16$$

$$\times 2^0 = 8$$

$$\checkmark 2^2 = 4$$

$$\text{loop} = 1 \text{ টি}$$

$$\therefore F = \bar{B} \bar{D}$$

This can also be done by,

MSB_R AB

	00	01	11	10
00	1 ₀	1 ₄	1 ₁₂	0 ₈
01	1 ₁	1 ₅	1 ₁₃	0 ₉
11	0 ₃	0 ₇	1 ₁₅	1 ₁₁
10	0 ₂	0 ₆	1 ₁₄	1 ₁₀

$$2^3 = 8$$

$$2^2 = 4$$

$$2^1 = 2$$

$$2^0 = 1$$

$$F = \bar{A}\bar{C} + B\bar{C} + CA$$

MSB_R AB

	00	01	11	10
00	1 ₀	1 ₄	1 ₁₂	0 ₈
01	1 ₁	1 ₅	1 ₁₃	0 ₉
11	0 ₃	0 ₇	1 ₁₅	1 ₁₁
10	0 ₂	0 ₆	1 ₁₄	1 ₁₀

$$F = \bar{A}\bar{C} + AB + CA$$

MSB_R AB

	00	01	11	10
00	1 ₀	0 ₄	0 ₁₂	1 ₈
01	0 ₁	1 ₅	1 ₁₃	0 ₉
11	0 ₃	1 ₇	1 ₁₅	0 ₁₁
10	1 ₂	0 ₆	0 ₁₄	1 ₁₀

$$F = BD + \bar{B}\bar{D}$$

$$= B \oplus D$$

Five variable k-Map :

- $2^5 = 32$
- ~~X~~ 1x32
 - ~~X~~ 2x16
 - ~~X~~ 4x8
 - ~~X~~ 8x4
 - ~~X~~ 2x16
 - ~~X~~ 32x1

calculated by using two Four-variable k-map:

		DE			
		00	01	11	10
BC	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

A=0

		DE			
		00	01	11	10
BC	00	16	17	19	18
	01	20	21	23	22
	11	28	29	31	30
	10	24	25	27	26

A=1

$$F(A, B, C, D, E) = \sum (0, 16)$$

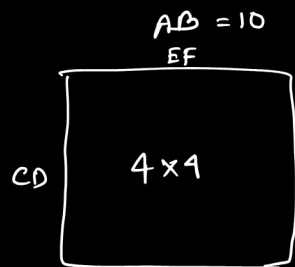
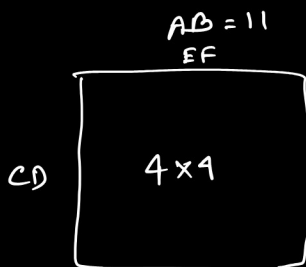
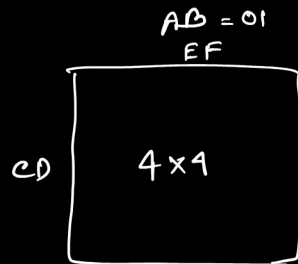
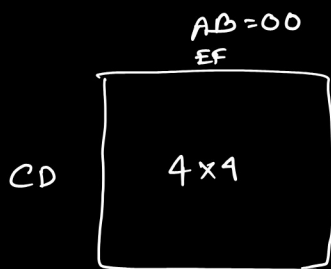
$$= \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + A\bar{B}\bar{C}\bar{D}\bar{E}$$

$$= \bar{B}\bar{C}\bar{D}\bar{E}(\bar{A} + A)$$

$$= \bar{B}\bar{C}\bar{D}\bar{E}$$

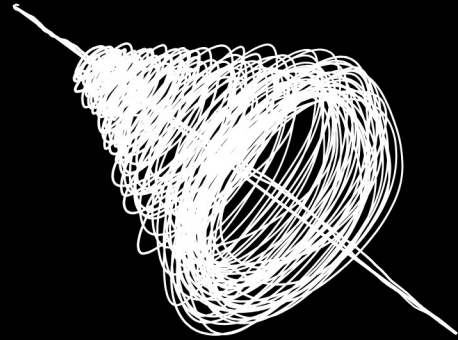
As one variable is eliminated, so 0 and 16 are adjacent

Six-Variable k-map:



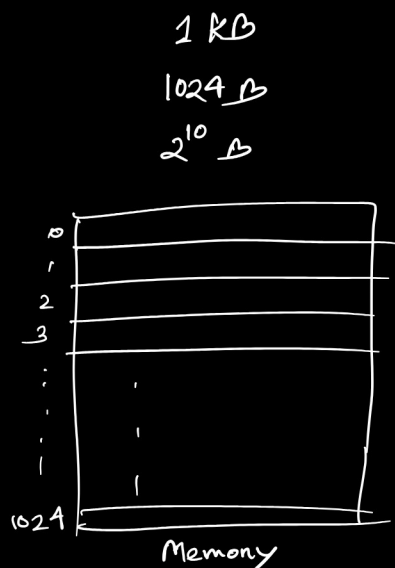
why the correction factor is 6 (0110) in BCD addition?

How?
 $4 \div 2$
 Circuit Design

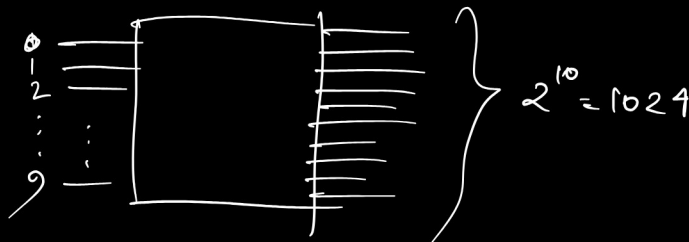


Use of Decoder:

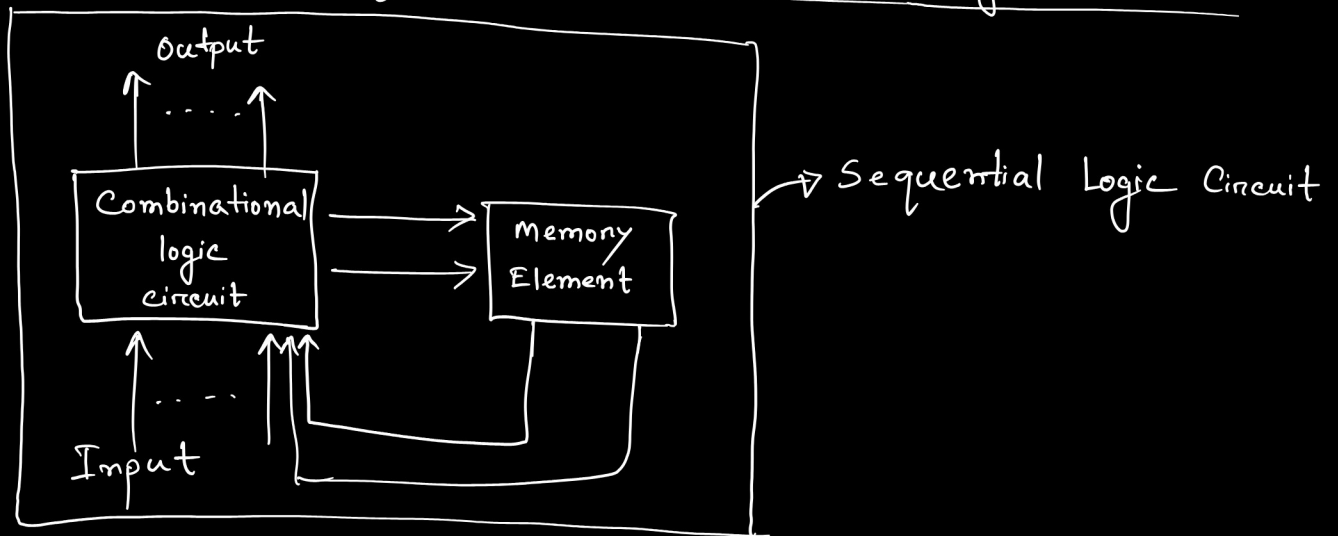
i) Memory management



Normally, 1024 2^r line 2^{10} line 2^{10} Decoder 2^{10} line
 10 2^r line 2^{10} line 2^{10} represent 2^{10} line.



Sequential Logic Circuit vs. Combinational logic circuit

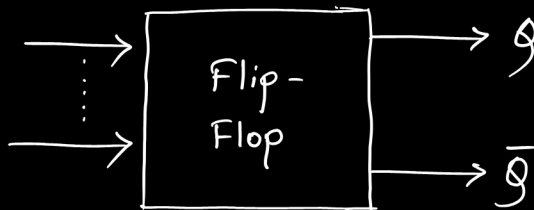


Main difference is memory element

Smallest memory unit is Flip-Flop (FF)

Flip-Flop

- can store 1-bit information (0 or 1).



4 types :

- i) SR
 - ii) JK
 - iii) D
 - iv) T
- } 2-input
- } 1-input

- 2-Input type - SR Flip-Flop
2 inputs are J and K, output
are 2 bits - Q and $\bar{Q}$
- Input 1 bit-3 ହେଉ ପାରେ,
2 bit-3 ହେଉ ପାରେ।

$Q = 0$ (Reset/clear) / Low

$Q = 1$ (Set) / High

Flip-Flop construction :

i) using NAND gates

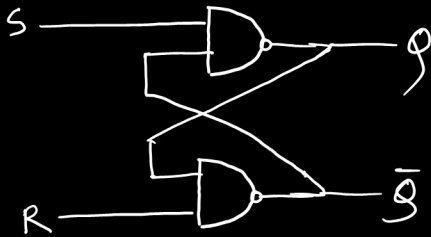
ii) using NOR gates

SR Flip-Flop using NAND gates

Block Diagram :



Logic Circuit :

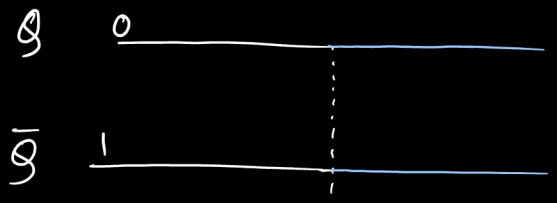
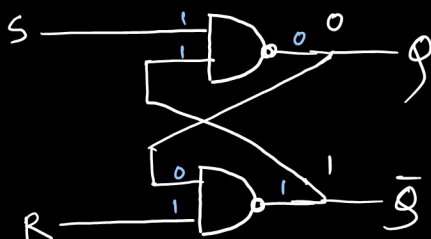


SR FF (NAND)

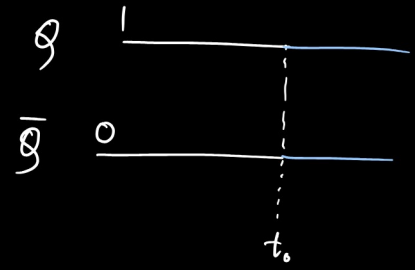
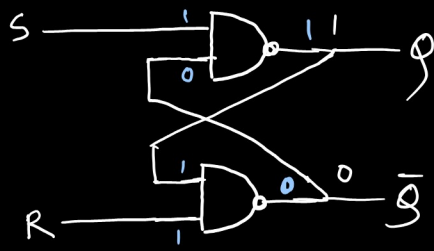
Truth Table

Case-1 : $S = 1$ and $R = 1$

a) Assume $Q = 0$, $\bar{Q} = 1$

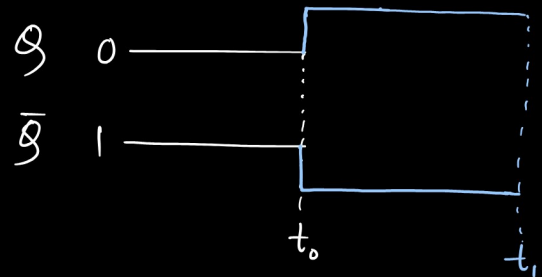
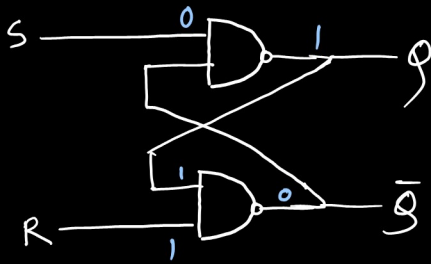


b) Assume $Q=1$, $\bar{Q}=0$

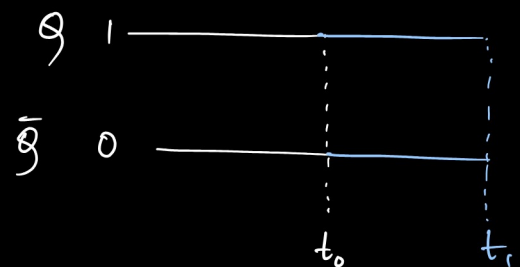
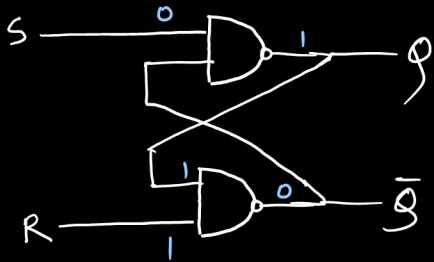


Case-2: $S=0$ and $R=1$

a) $Q=0$, $\bar{Q}=1$

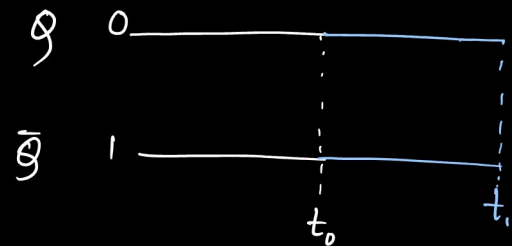
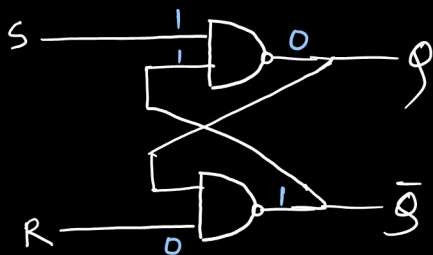


b) $Q=1$, $\bar{Q}=0$

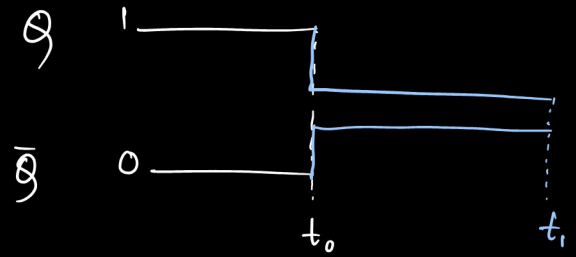
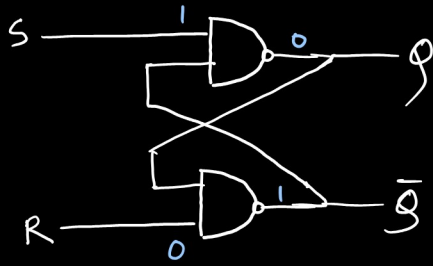


Case-3: $S=1$ and $R=0$

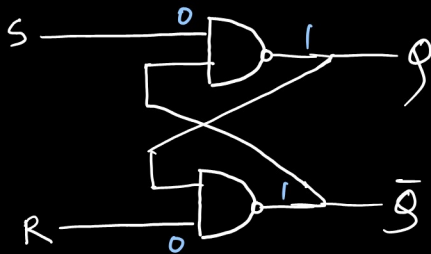
a) $Q=0$, $\bar{Q}=1$



b) $Q=1, \bar{Q}=0$



Case-4: $S=0$ and $R=0$



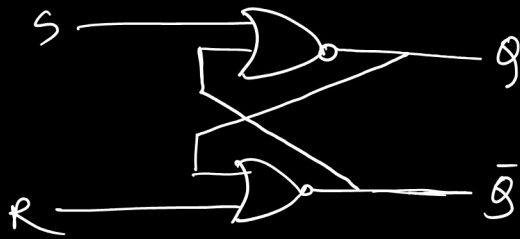
Here, $Q=\bar{Q}=1$

But this is not possible. So, this case is invalid.

Truth Table:

S	R	Q
1	1	No change/ Memory/ Copy
0	1	1 (set)
1	0	0 (Reset)
0	0	Invalid

SR Flip-Flop using NOR gates:



Truth Table:

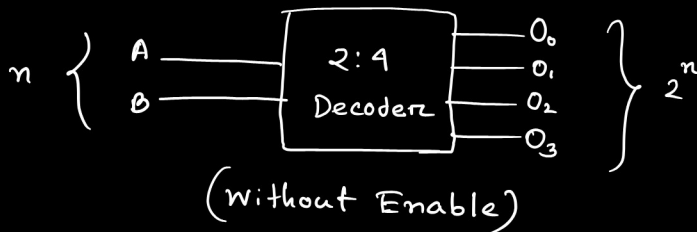
S	R	Q
1	1	Invalid ($Q = \bar{Q} = 0$)
0	1	Reset (0)
1	0	Set (1)
0	0	No change

Analysis (Homework)

Class-23

25/08/26

Decoder



(without Enable)

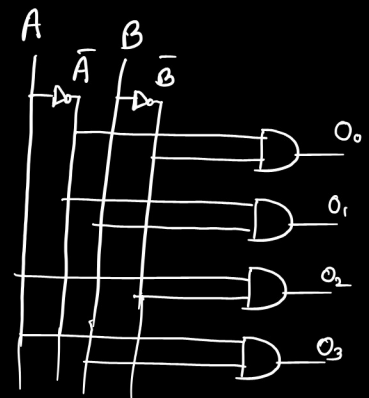
Input		Output			
A	B	Q_0	Q_1	Q_2	Q_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

$$Q_0 = \bar{A}\bar{B}$$

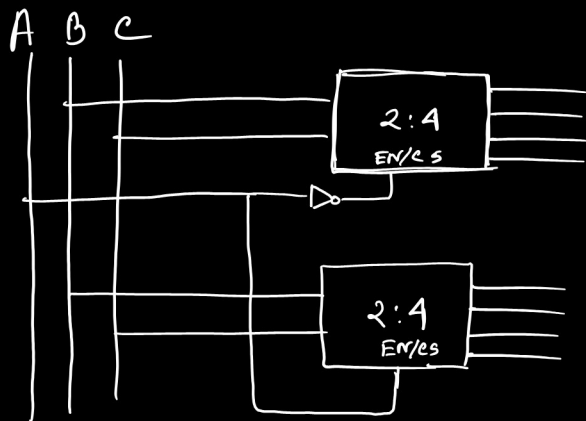
$$Q_1 = \bar{A}B$$

$$Q_2 = A\bar{B}$$

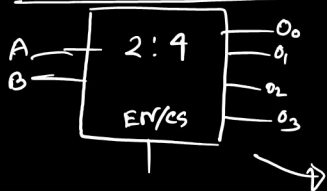
$$Q_3 = AB$$



3:8 Decoder using 2:4 Decoder:



2:4 Decoder with Enable:



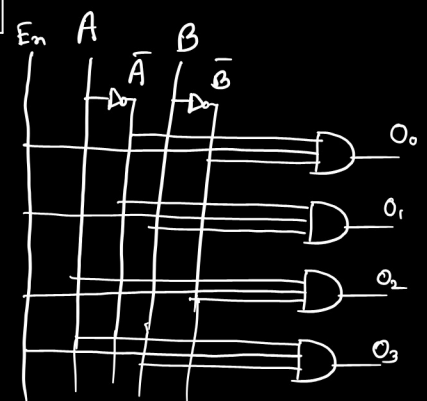
Input			Output			
E_n	A	B	O_0	O_1	O_2	O_3
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

$$O_0 = \bar{A} \bar{B} E$$

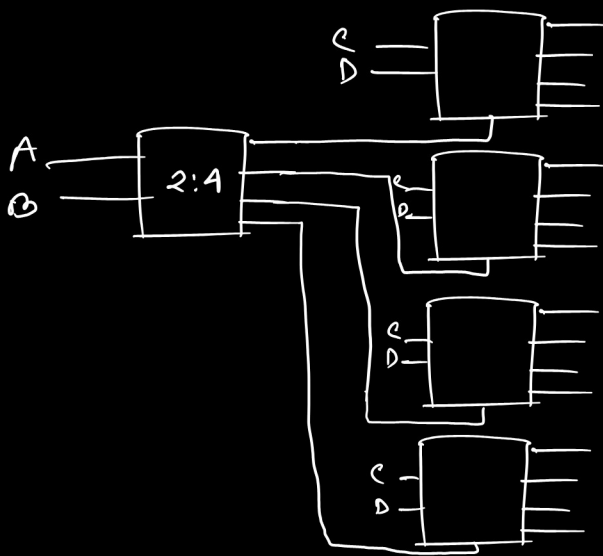
$$O_1 = \bar{A} B E$$

$$O_2 = A \bar{B} E$$

$$O_3 = A B E$$

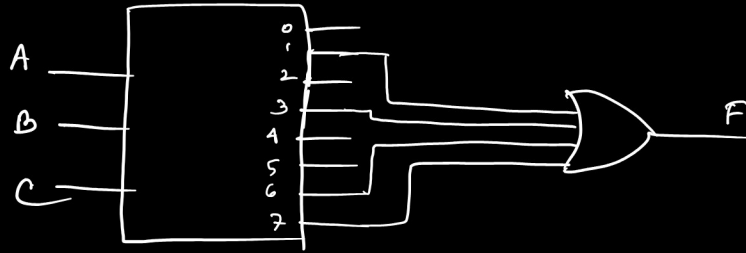


4:16 Decoder using 2:4 Decoder:



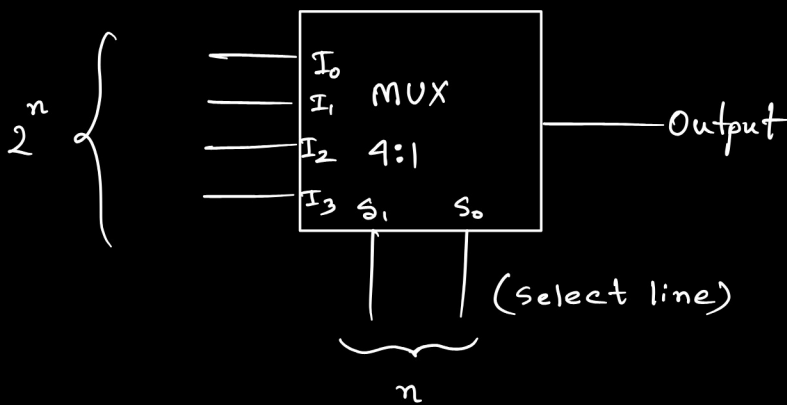
$F(A, B, C) = \sum(1, 3, 6, 7)$ Design using Decoder. [Specific কোনাে Decoder বানানো গেলে use করতে হবে]

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



Designing Full adder/Half adder using Decoder

Multiplexer :



* Works as a switch

Application :

- Telephone exchange
- PBX
- Intercom Telephone
(Building-এর resident-এর telephone-এর
এবং guard-এর telephone-এর
মধ্যে communication)

• At a time, output -এরটো input -এর মাঝে
যোগ হবে।

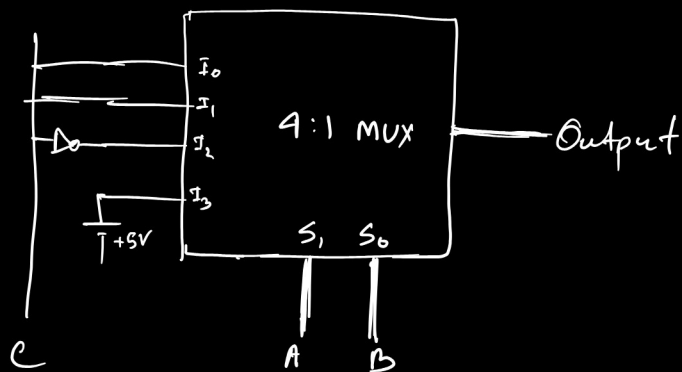
Input	Select Line	Output
I	S_1, S_0	F
I_0	0, 0	I_0
I_1	0, 1	I_1
I_2	1, 0	I_2
I_3	1, 1	I_3

$$\therefore F = I_0 \bar{S}_0 \bar{S}_1 + I_1 \bar{S}_0 S_1 + I_2 S_0 \bar{S}_1 + I_3 S_0 S_1$$

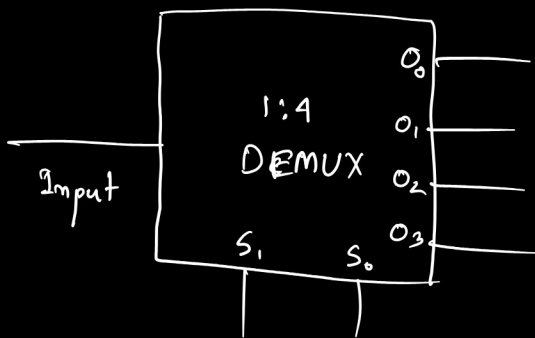
$F(A, B, C) = \Sigma(1, 3, 4, 6, 7)$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

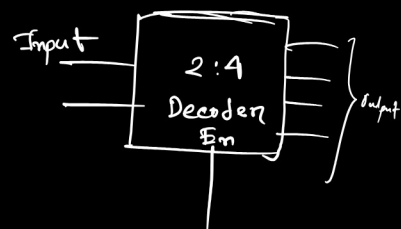
$\rightarrow F = C$
 $\rightarrow F = C$
 $\rightarrow F = \bar{C}$
 $\rightarrow F = 1$



Demultiplexer:

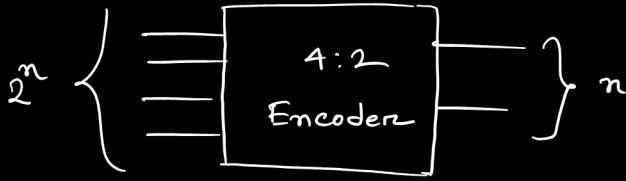


It is identical to



So, Decoder with enable is also called demultiplexer.

Encoder :

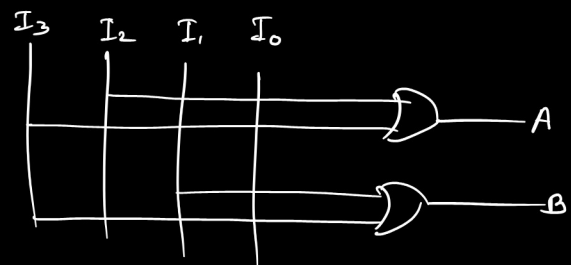


Input				Output	
I_3	I_2	I_1	I_0	A	B
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

16
Combinations
?

?

$$A = \bar{I}_0 \bar{I}_1 I_2 \bar{I}_3 + \bar{I}_0 \bar{I}_1 \bar{I}_2 I_3 = I_2 + I_3$$
$$B = \bar{I}_0 I_1 \bar{I}_2 \bar{I}_3 + \bar{I}_0 I_1 I_2 \bar{I}_3 = I_1 + I_3$$



Priority Encoder (self-study)